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Palaeomagnetic evidence for the origin of the coastal flexure and dyke swarm in central E Greenland

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SUMMARY: The well known coastal flexure of central E Greenland, with its associated dyke swarm, affects a thick pile of Tertiary basalts and the underlying basement gneisses. It is a major tectonic/magmatic feature of this 'trailing edge' type continental margin. The original interpretation of Wager & Deer (1938) envisaged the emplacement of the dykes as a fan-shaped swarm in the convex arc of the flexure. Nielsen (1975, 1978) recognized several episodes of dyke emplacement. He interpreted the main swarm (THOL 1) as earlier than the flexure, emplaced vertically into the horizontal basalt pile and subsequently tilted by antithetic faulting during the formation of the flexure. A palaeomagnetic study of the THOL 1 and later dykes shows that the former were emplaced in their present inclined attitude. This discriminates in favour of Wager's interpretation, while remaining consistent with Nielsen's view of the post-flexural magmatic history. Modifications to Nielsen's scheme are suggested to conform with the new evidence, and the implications in relation to the mechanism of flexuring, the timing of tholeiitic magmatism, and the relationship of both to the early spreading history of this sector of the NE Atlantic, are discussed.

The coastal flexure in E Greenland is a seaward facing monocline which marks the present coastline for some 600 km between approximately 66 and 70°N (Fig. 1). To the NE of Kangerdlugssuaq, along the Blosseville Kyst, the flexure affects Palaeogene basalts. On the outer headlands for 75 km as far as Jensen Fjord, seaward dips of 35–55° are observed and the monocline is associated with an intense swarm of basic dykes which dip landward, roughly normal to the flows. Further NE the coast recedes. Dips are shallow and the dyke swarm less intense; the flexure apparently lies offshore. Towards the northern end of the Blosseville Kyst seaward dips of up to 30° again occur but these are in part due to block faulting and the flexure is weak. To the SE of Kangerdlugssuaq, at a deeper erosional level, the course of the flexure is marked in the basement gneiss complex by en echelon zones of intense dyke injection (Bridgwater *et al.* 1978). The dykes dip landward and, where patches of extrusive basalt are preserved, are again normal to the flows.

Since their discovery (Wager & Deer 1938; Wager 1947) the dyke swarm and flexure have come to be regarded as a classic example of extensional-type magma tectonics on a continental margin. Wager's interpretation of the structure presaged modern ideas on the association of vertical displacements with changes in crustal thickness and thermal conditions in the mantle. He viewed the dykes as a fan-shaped swarm emplaced in the antiformal (convex) part of the monocline due to outer-arc extension (Fig. 2A). This has been questioned by Nielsen (1975), who proposed that the major set of coast-parallel tholeiite dykes was emplaced vertically, into horizontal basalt flows, as a regional swarm associated with pre-spreading uplift and doming (Brooks 1973). Antithetic block faulting during the main rifting episode then tilted the flows seaward so that the dykes dip landward, producing the apparent fan configuration (Fig. 2B). Later dyke sets were emplaced steeply in the already tilted blocks and mark further extension during early spreading. As a modification to this hypothesis, Nielsen (1978) suggested that the 'pre-flexure' dykes (THOL 1 in his petrologically and structurally defined sequence) curve over inland to form the extensive sill complex E of Kangerdlugssuaq (Figs. 1, 2C). The steep part of the swarm supposedly marks the early spreading axis and acted as feeder to the chemically comparable plateau basalts. This idea was adopted by Larsen (1978) in his interpretation of aeromagnetic data from the region.

In 1977 the authors visited the Jacobsen Fjord area with Dr Nielsen and, as part of a larger palaeomagnetic study, sampled the early and later dykes under his guidance. At Kap Irminger (Figs. 1, 6D) the THOL 1 dykes dip landward at 45–60°, are often more than 10 m in thickness and occupy about 60% of the outcrop. Screens of country rock basalt and tuff make up about 35%. Post-flexure dykes are fewer (5%), thinner and subvertical—precisely as described by Nielsen (1978, fig. 2) from Haengefjeldt at Kangerdlugssuaq. However, in inland sections we were unable to verify that the early dykes pass into the sill complex, or that the coast-parallel faulting, which is certainly present, is responsible for the flexure. Clearly, it is possible in principle to discriminate palaeomagnetically between dykes which were emplaced as inclined sheets, as envisaged by Wager, and vertical dykes which were subsequently tilted, as in Nielsen's scheme.

Palaeomagnetic investigation

Experimental procedure

Around the head of Jacobsen Fjord (68°14′N, 31°00′W; Fig. 1) one site was drilled in each of 44 basalt flows. Details of this study will be re-

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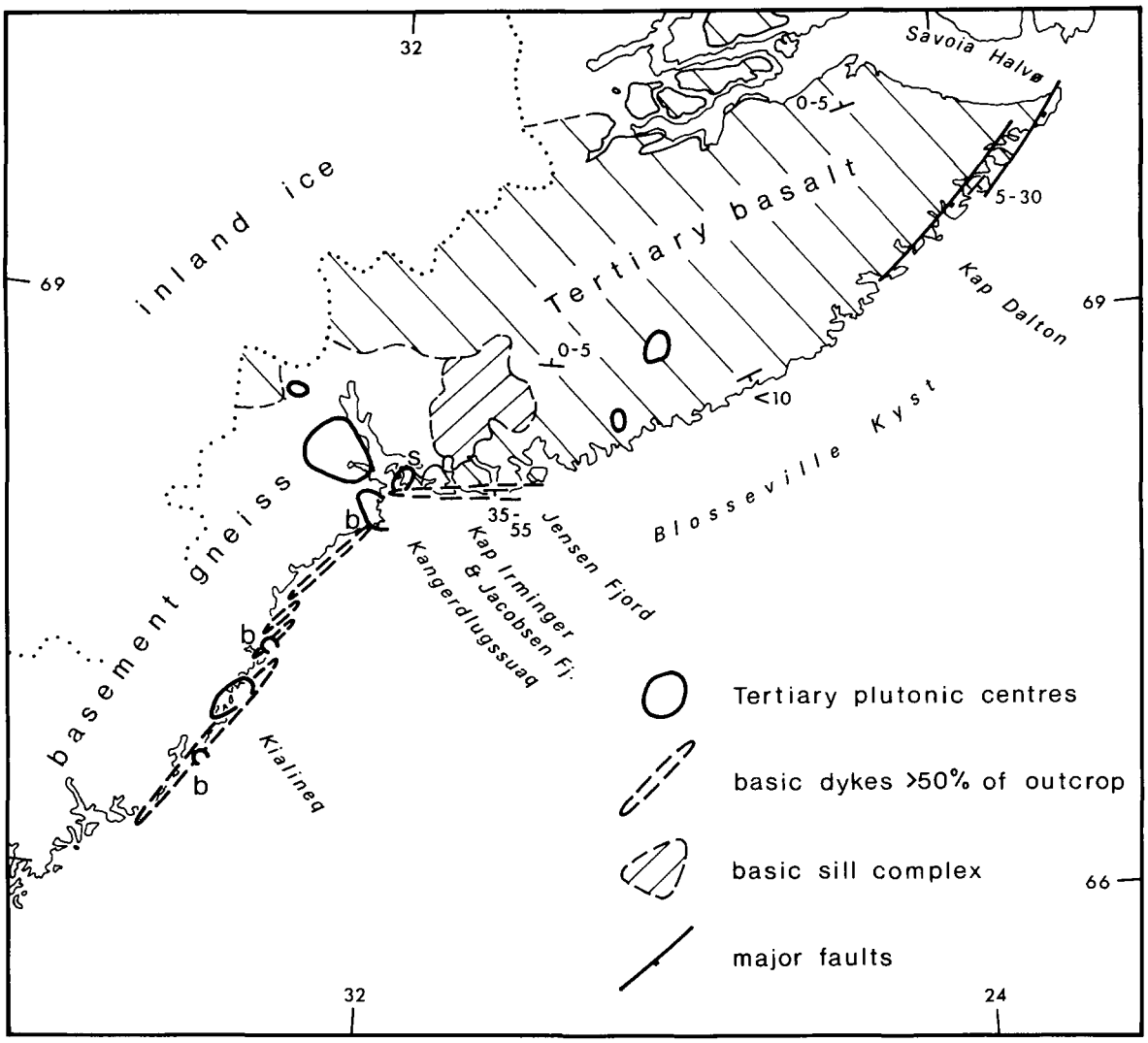


FIG. 1. Outline Tertiary geology of central E Greenland, partly after Bridgwater *et al.* (1978), showing places mentioned in the text. S: Skaergaard intrusion; b: outlying patches of extrusive basalt.

ported elsewhere. Additionally, at sites along the fjord sides, 22 younger (post-flexure) dykes were sampled and 11 THOL 1 dykes were sampled at Kap Irminger along an approximately 200 m section. This contained 5 younger dykes of which 3 (sites 96, 97 and 98) were sampled. The only non-dyke site (73) at Kap Irminger was in a screen of feldspar and pyroxene-phyric basalt, 0.5 m from where a THOL 1 dyke chills against it. 6 cores were drilled at each site and oriented by sun compass.

Palaeomagnetic measurements were made on a *Digico* balanced fluxgate spinner magnetometer. One specimen from each site was subjected to step-wise thermal demagnetization and another to step-wise alternating field (AF) demagnetization using a two-axis tumbler. The lowest peak field at which the pilot specimen showed stable directions was used to treat the remaining 4 specimens from each site. Finally one of these 4 specimens, from which any low coercivity component had been removed by the AF method, was progressively thermally demagnetized by heating to temperatures around 600°C, in order to examine the blocking temperature spectrum of the stable component. Thermal demagnetization was carried out in a field of less than 5 gamma in an atmosphere of oxygen-free nitrogen. Initial susceptibilities were measured on all cores. In addition to the Jacobsen Fjord collection measurements were made on 3 cores from a hand sample of trough band G in the Skaergaard intrusion (Wager & Deer 1939) (kindly supplied

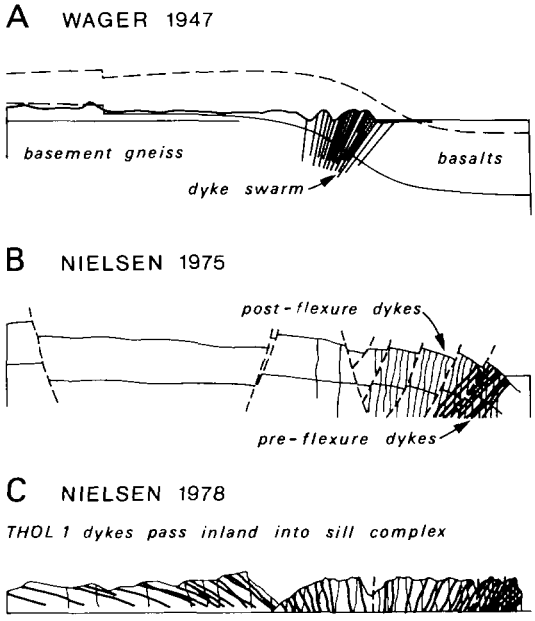


FIG. 2. Previous interpretations of the coastal flexure and dyke swarm.

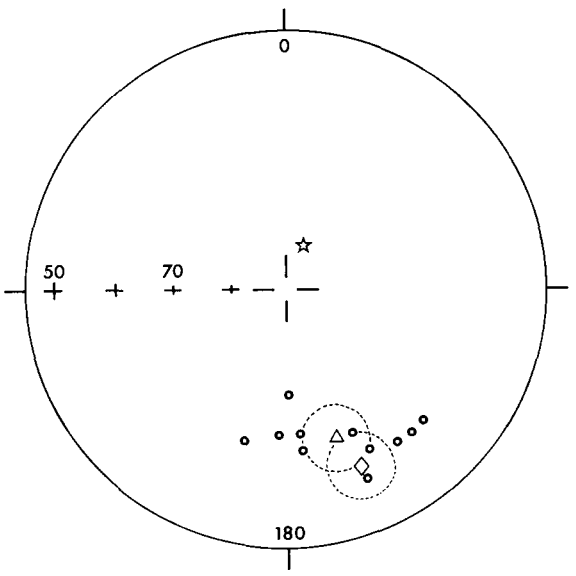


FIG. 3. Stereographic projection of stable palaeomagnetic directions. Open symbols denote the upper hemisphere (reversed). Circles represent mean *in situ* directions for THOL 1 dykes. The triangle is the overall *in situ* mean direction for the THOL 1 dykes with its α_{95} shown. The star is the corresponding overall mean if the dykes are tilted to vertical. The diamond is the stable tilt-corrected overall mean for basalt flows from the Jacobsen Fjord area, with its α_{95} shown. Note that in Figs. 3, 4 and 5 the primitive corresponds to an inclination of 45°.

by Professor P. E. Brown), in order to check the polarity of the remanence.

Results

The mean palaeomagnetic direction from 39 basalt flows is $D = 157°$, $I = -56°$, $\alpha_{95} = 6°$ after demagnetization to a stable end point and correction for tilting. This mean direction is close to that found from the Vandfaldsdalen Formation, $D = 159°$, $I = -63°$, $\alpha_{95} = 9°$ (Faller 1975) and to those from the Scoresby Sund area, $D = 168°$, $I = -62°$, $\alpha_{95} = 15°$ (Tarling 1967) and $D = 165°$, $I = -65°$, $\alpha_{95} = 5°$ (Hailwood 1977).

The THOL 1 dykes all have stable reversed inclinations and low within-site dispersion (Table 1, Fig. 3). Their mean stable *in situ* direction is $D = 161°$, $I = -62°$, $\alpha_{95} = 6°$, which is statistically identical to the mean corrected direction for the Jacobsen Fjord basalts according to the test of Watson (1956). They all have high remanence intensity and high initial susceptibility, which together give Königsberger ratios close to unity in most cases (Table 1). An unstable normal component of magnetization is often present initially; this is removed by AF demagnetization at peak fields below 5 mT. The reversed component has high directional stability after heating to temperatures approaching the Curie point of magnetite (578°C) (Fig. 5). The blocking temperature spectrum of all the THOL 1 dykes extends up to the Curie point of magnetite with an appreciable fraction of the intensity remaining up to the highest temperature (Fig. 5). The flow intruded by THOL 1 dykes (site 73) is magnetized in a similar direction to these dykes (Table 1) but a specimen from it moves to shallower inclinations on heating to 560°C (Fig. 5).

The post-flexure dykes include 15 with normal polarity stable inclinations (mean direction $D = 352°$, $I = 66°$, $\alpha_{95} = 4°$) and 7 which are reversed ($D = 150°$, $I = -62°$, $\alpha_{95} = 7°$) (Table 2, Fig. 4). The Skaergaard cores have extremely high remanence intensities and high initial susceptibilities, as would be expected from the high modal content of magnetite in the mafic layers. They are reversely magnetized and directionally stable (Table 1).

Interpretation

Our data on the Königsberger ratios of the THOL 1 dykes are relevant to the interpretation of magnetic anomalies presumed to be associated with the presence of this dyke swarm offshore (Larsen 1978). Most of these ratios are close to unity, which implies that the magnetization induced by the present Earth's field is comparable to the remanence. Both magnetizations

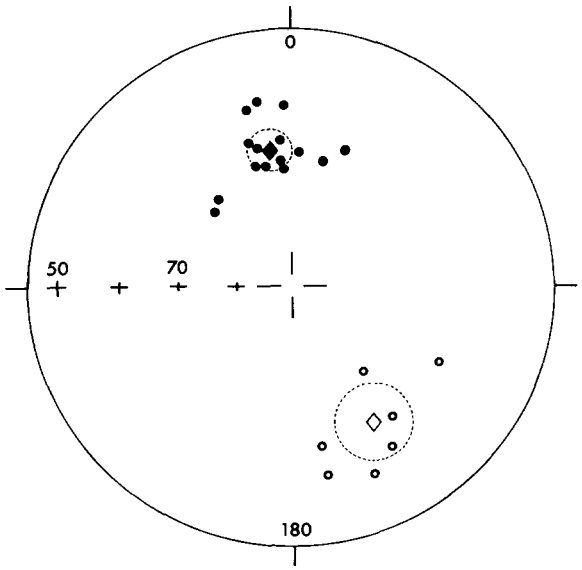


FIG. 4. Stereographic projection of stable *in situ* directions for post-flexure dykes. Open symbols denote the upper hemisphere (reversed) and solid symbols the lower hemisphere (normal). Circles refer to individual dykes and diamonds to the overall means of normal and reversed dykes respectively. The α_{95} of each group mean is shown.

would therefore contribute to a magnetic anomaly. In the dykes with reversed initial inclinations the remanence and the induced component tend to oppose each other and would produce a smaller anomaly than either effect alone.

The mean stable *in situ* direction for the THOL 1 dykes is in very close agreement with the mean direction from the basalt flows after they have been restored to the horizontal. If a tilt correction is applied to the THOL 1 dykes, as though they had been intruded vertically and tilted subsequent to magnetization, the inclination becomes very steep ($D = 027°$, $I = -81°$) and the mean direction no longer agrees with that from the basalts (Fig. 3).

3 possibilities must be considered. The first is that the THOL 1 dykes were intruded vertically at a time when the magnetizing field was appropriate to steep inclinations and tilted to their present position afterwards. However, it is clear that the THOL 1 dykes were intruded very soon after the eruption of the flows, probably in the same polarity epoch (see below), so that there was insufficient time for an appreciable change in the field between the intrusive and extrusive activity. Comparison of the 'tilt corrected' directions for the dykes with the directions for individual flows suggests that the steep inclinations are unlikely to arise from secular variation. Furthermore, the steep inclinations do not correspond to any known Tertiary pole

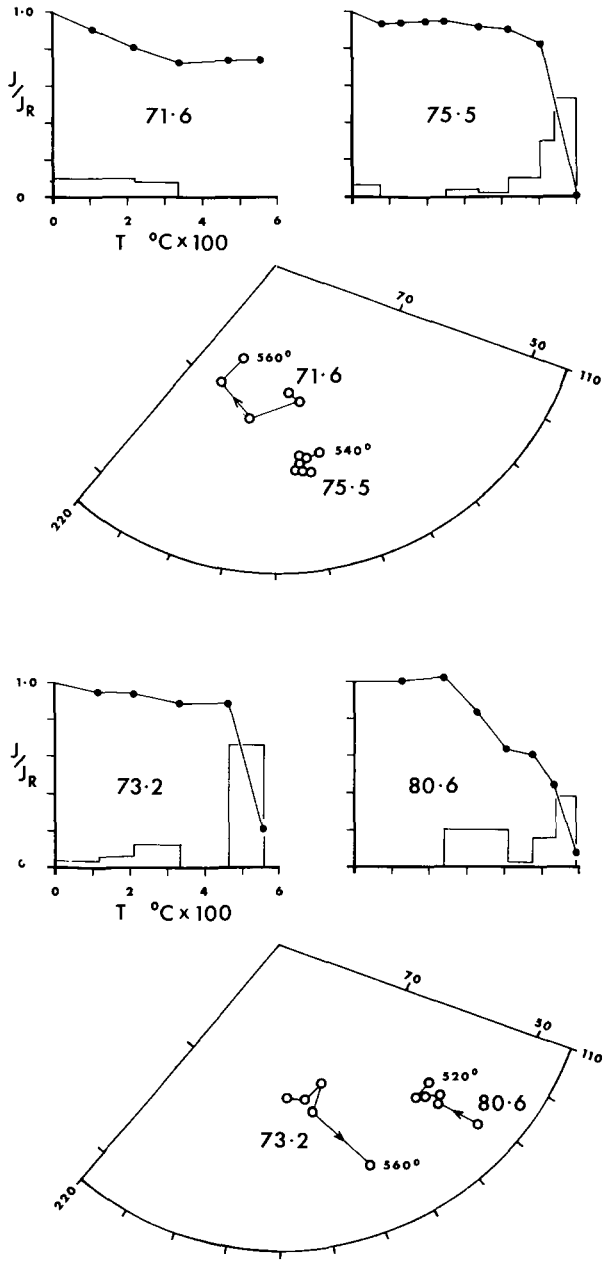


FIG. 5. Examples of thermal demagnetization after removal of any low coercivity component by AF cleaning. 71.6, 75.5 and 80.6 show typical behaviour of specimens from THOL 1 dykes; 73.2 is from the flow intruded by THOL 1 dykes. The proportion of the stable remanence intensity removed in each temperature interval is indicated below the intensity decay curves. The final temperature reached is shown on the stereographic projections.

TABLE 1: *Palaeomagnetic results from THOL 1 dykes etc. (1 site per dyke)*

Site	*Dip*	*Strike*	*N*	$\bar{J}_0$	$\bar{X}$	$\bar{Q}$	D_0	I_0	α_{95}	D	I	α_{95}	MDF	D_v	I_v
69	52	092	6	1.9	0.047	0.9	209	−11	15	196	−62	6	6	330	−78
70	53	095	6	4.0	0.052	2.0	188	−48	8	175	−64	5	6	026	−77
71	54	098	6	1.6	0.032	1.3	201	−35	11	179	−71	4	20	018	−73
72	60	094	6	7.4	0.044	4.5	195	−57	5	183	−64	4	20	012	−86
74	54	086	6	2.3	0.040	1.4	166	−40	7	156	−62	5	15	042	−77
75	60	090	6	18.7	0.030	15.2	179	−59	4	175	−61	3	12	067	−87
76	54	085	6	1.6	0.054	0.8	166	−21	16	153	−58	6	12	057	−77
77	58	093	6	1.1	0.024	1.1	143	+1	12	144	−57	3	21	078	−70
78	62	081	6	1.2	0.052	0.6	159	−11	17	157	−54	3	9	124	−79
79	46	083	6	1.2	0.044	0.7	149	+44	7	139	−57	4	45	042	−67
80	53	107	6	2.2	0.014	2.7	131	−48	4	134	−57	2	38	074	−55
Overall Mean			11	3.8	0.038					161†	−62†	6		027	−81
73	Basalt Flow			2.7	0.022	3.1	178	−64	4	178	−66	4	35		
	Skaergaard Intrusion		1	10.2	0.035	7.3		−55*			−55*		15		
			1	81.7	0.120	16.6		−27*			−37*		25		
			1	19.3	0.040	12.0		−50*			−49*				

*Assuming layering to be horizontal. †Corresponding palaeomagnetic pole position; 180.4 E, 62.9 N.
Key to palaeomagnetic symbols: N = number of samples; $\bar{J}_0$ = mean NRM intensity (Am^{-1}): $\bar{X}$ = mean initial volume susceptibility (SI units); $\bar{Q}$ = mean Königsberger ratio; D_0 = initial declination; I_0 = initial inclination; D = stable declination; I = stable inclination; α_{95} = semi-angle of cone of 95% confidence of preceding direction; MDF = median destructive field (mT); D_v = declination on tilting to vertical; I_v = inclination on tilting to vertical.

TABLE 2: *Palaeomagnetic results from post-flexure (dykes (1 site per dyke).*

Site	*N*	$\bar{J}_0$	D_0	I_0	α_{95}	D	I	α_{95}
22	6	0.1	161	−42	27	157	−54	16
81	5	1.7	144	−52	3	149	−57	2
82	5	2.1	014	+63	5	022	+64	7
83	5	2.0	353	+64	2	357	+69	5
84	5	2.2	175	−62	2	170	−66	3
85	5	0.2	353	+66	3	356	+64	4
86	5	18.7	350	+59	3	350	+57	3
87	5	3.1	007	+59	7	358	+58	8
88	5	3.8	348	+69	4	347	+65	6
89	5	6.4	006	+68	5	004	+68	6
90	5	7.1	349	+60	2	346	+58	1
91	5	5.2	143	−66	4	141	−70	5
92	5	5.4	351	+56	4	348	+68	3
96*	5	1.3	Not Significant			118	−61	4
97*	5	4.6	173	−54	4	170	−61	3
98*	5	1.2	130	−44	4	144	−61	5
99	5	3.6	045	+78	2	015	+67	5
100	5	2.2	354	+66	5	345	+64	3
101	5	8.4	344	+68	2	345	+68	3
102	5	3.5	328	+72	5	320	+70	3
103	5	0.3	327	+72	4	315	+71	3
104	5	0.2	353	+71	3	357	+68	9
	Mean of normal dykes					352	+66	4
	Mean of reversed dykes					150	−62	7

* Sampled at Kap Irminger.

position. We therefore eliminate this possibility and consider only the *in situ* directions of magnetization.

The remaining possibilities are that the THOL 1 dykes were intruded and magnetized in the present inclined attitude or that we are seeing the effects of remagnetization, presumably associated with the intrusion of the post-flexure dykes. Although 3 younger dykes at Kap Irminger have *in situ* directions close to the mean of the THOL 1 dykes, we do not consider that the *in situ* THOL 1 directions are due to remagnetization, for the reasons discussed below.

Sites 72, 74, 77, 79 and 80, which were located within a few metres of younger dykes, show the same high directional stability as the other THOL 1 sites, which were drilled some tens of metres from the nearest younger intrusion. In particular, THOL 1 sites 77, 79 and 80 are close to younger dyke sites 96, 98 and 97 respectively; 77 and 96 differ directionally, as do 80 and 97.

If the THOL 1 dykes had been intruded vertically, magnetized in the same field as were the flows they intrude, then tilted to dip some 30 or 40° northward, their *in situ* directions would be appreciably shallower than are observed. However, none of these dykes moves to shallower inclination on thermal demagnetization, as would be expected if the observed *in situ* directions are the effect of partial remagnetization. On the other hand, the basalt flow intruded by the THOL 1 dykes (site 73) does move to shallower inclination after heating to 560°, as though it is reverting to the stable remanence acquired when it cooled prior to being tilted. It appears, then, that this flow was no longer horizontal when it was remagnetized by the intrusion of the THOL 1 dykes. Since the intrusion of the massive THOL 1 dykes did not completely mask the primary remanence of the pre-existing flows, it is

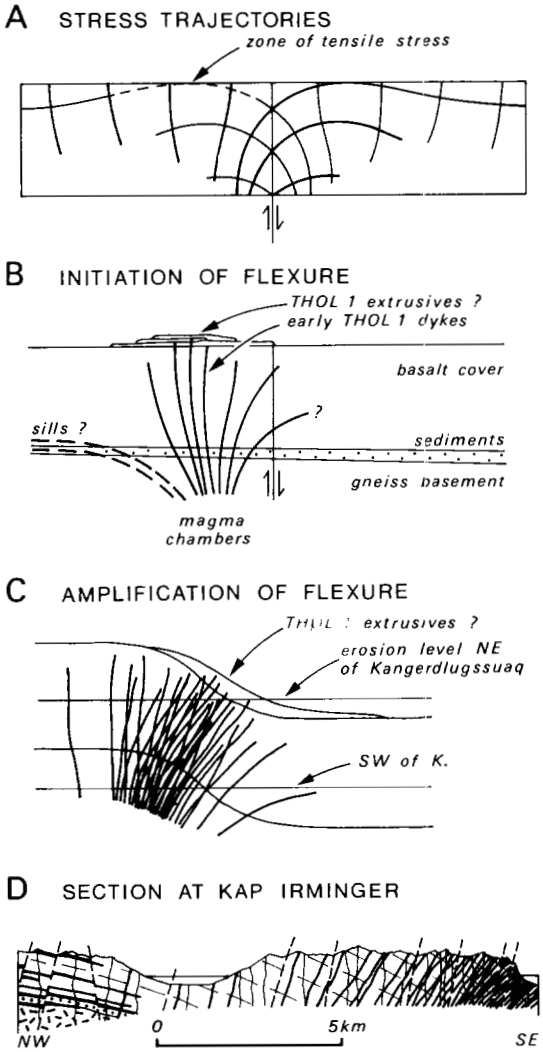


FIG. 6. Interpretation of the coastal dyke swarm and flexure. A: Conjugate principal stress trajectories induced in an elastic cover overlying a small step-like displacement in the basement, after Sanford (1959). B: The model applied to the E Greenland situation. Dykes with perhaps extrusives developed in the tensile zone. C: Amplification of the monoclinal flexure by differential vertical displacement, accommodated by extensional dyke emplacement. D: Schematic section inland from Kap Irminger showing THOL 1 dykes (bold), sills (bold), later dykes (thin), sediments (dots) and basement (hatched).

most unlikely that the intrusion of the much thinner and fewer younger dykes could have completely remagnetized the THOL 1 dykes by heating them above their highest blocking temperatures.

We conclude that the THOL 1 dykes were intruded and magnetized in their present position. The similarity of the stable *in situ* directions of both the THOL 1 dykes and the post-flexure dykes to the tilt-corrected directions of the basalts confirms that the magnetizing field changed little between the extrusion of the basalt pile and the intrusion of the latest dykes, although this interval could be as much as 15 Ma (see below).

Discussion and conclusions

Mechanism

We have shown that the THOL 1 dykes at Kap Irminger cooled in approximately their present position and that their magnetization is in all probability a primary thermo-remanence. It follows that the dykes were not emplaced vertically and subsequently tilted, but were formed in much their present position in association with the development of the coastal flexure. This confirms Wager's interpretation, which was based on the spatial association of the dyke swarm with the flexure, and the manner in which the densest part of the swarm coincides with the steepest part of the monocline.

Criticism of Wager's interpretation has centred on the problem of transmitting basaltic liquid across a compressional zone in the inner arc of the antiformal part of the flexure. This problem arises only if the mechanism is one of layer-controlled buckling. We envisage a bending mechanism involving differential vertical displacement of the crust in which the basalt cover acted passively. Orientation of the dykes was presumably controlled by the stress pattern induced in the upper crust by this bending. Sanford's (1959) analysis of stress trajectories within an elastic material representing the cover overlying a step-like displacement in the basement, shows sub-horizontal tensile stress to be developed in the antiformal part of the incipient flexure (Fig. 6A). The model predicts a concentration of dykes in that position (Fig. 6B) and allows two interesting possibilities. One is that the deeper level intrusions turned over seawards to form low-angled sheets, the sense of curvature being opposite to that recorded by Nielsen (cf. Figs. 6B, 2C). The introduction of voids into the model, to represent magma chambers, would no doubt allow the possibility of landward curving sheet intrusions (shown schematically on Fig. 6B). The second possibility is that some of the dykes reached the surface to form fissure eruptions along the line of the flexure. Evidence from Kap Dalton (admittedly 300 km to the NE, where the dyke swarm is weaker) shows that there the dykes did not penetrate the upper part of the basalt pile (Watt *et al.* 1976). In the Kangerdlugssuaq region it is clear that the THOL 1 dykes do not represent feeders to the chemically comparable plateau basalt flows which they

penetrate. Since the whole pile thickens seaward it is thought to have been erupted from fissure systems further offshore (Soper *et al.* 1976*b*). However, the possibility remains that the THOL 1 dykes fed flows now removed by erosion, but perhaps preserved at the top of the eruptive pile offshore (Fig. 6C).

Sanford's analysis is based on a very small vertical displacement (about one five hundredth of the thickness of the cover) and is thus relevant only to the initiation of the flexure. Amplification of the structure to produce a monocline with a vertical displacement approaching the thickness of the basalt pile presumably resulted in the landward migration of the antiformal arc and the maintenance of approximately layer-parallel tensile stress in the developing common limb. We envisage the passive rotation of early formed dykes and continued emplacement of later dykes parallel to them to form the dense concentration on the common limb, the whole zone cooling as a unit towards the end of this period of activity (Fig. 6C).

Chronology

The palynologically determined age span of the basalt pile (Soper *et al.* 1976*a*, *b*) equates with the reversed geomagnetic polarity epoch 24–25, dated 52–55.5 Ma on the time scale of Hailwood *et al.* (in press). We have shown that emplacement of the THOL 1 swarm was associated with the flexuring, not eruption of the basalt pile, but evidence exists to show that these events were almost synchronous. The THOL 1 dykes are of FeTi type, chemically comparable with the plateau basalts (Nielsen 1978). They apparently pre-date the Skaergaard intrusion and its satellites, which are of MORB type (Nielsen 1978; Brooks & Nielsen 1978). Skaergaard has given a zircon fisson-track age of 54.6 ± 1.7 Ma (Brooks & Gleadow 1977). Reversed polarity has been reported for the intrusion (Girdler 1961) and is confirmed above. Is it possible that all this diverse tholeiitic activity took place within a single 3.5 Ma reversed epoch?

Acceptance of an age of *c.* 53 Ma for Skaergaard would just accommodate its emplacement into the basalt pile and subsequent tilting by the coastal flexure (Wager & Deer 1939) but not its post-THOL 1 age. The intrusion undoubtedly is tilted (although cut by only post-flexure dykes) but not necessarily as much as the enclosing basalts. Nielsen (1978) observed pre-consolidation disruption of cumulous layering in the Sødalen macrodyke, an offshoot of Skaergaard, and fig. 2 of his paper showed the Skaergaard-related sill on Haengefjeldt to be cut by THOL 1 dykes. It appears quite possible that Skaergaard, together with its attendant macrodykes and the THOL 2 radial swarm (and also the extensive tholeiitic sill complex E of Kangerdlugssuaq, Fig. 1), were all emplaced synchronously with the flexuring at *c.* 52–53 Ma.

If this is true, there is an interesting corollary regarding the early differentiation history of the Skaergaard intrusion. In an elegant chemical study of the Skaergaard-related THOL 2 (or FG-1) dykes Brookes & Nielsen (1978) confirmed that the large volume of early cumulates postulated in the earlier differentiation models must have been formed. Since geophysical investigations have failed to locate a 'hidden layered series' beneath the exposed intrusion Brooks and Nielsen proposed that early differentiation took place elsewhere, possibly beneath the outer part of Kangerdlugssuaq fjord at the focus of the THOL 2 radial swarm where a large gravity and magnetic anomaly exists. This is on the line of the flexure. Synchroneity of the flexuring and emplacement of Skaergaard would support this idea. It could have been the distortional effect of the flexuring on the early magma chamber which expelled the Skaergaard liquid towards its present position in the antiformal part of the flexure.

Geotectonic implications

A picture emerges of a burst of tholeiitic activity, both extrusive and intrusive and involving basaltic liquids of both FeTi and MORB type, in the period 52–55.5 Ma. This, as argued previously (Soper *et al.* 1976*a*,*b*), marks the initial rifting episode between central E Greenland and the Eurasian plate: the earliest oceanic anomaly (24) in the regions to the N and S is dated 51–52 Ma on the Hailwood scale. This brief tholeiitic phase was succeeded on the Greenland trailing edge by a period of extensional faulting and the emplacement of differentiated dykes of both alkaline and tholeiitic type (Nielsen 1978). At Jacobsen Fjord both groups are of mixed polarity, suggesting emplacement over an extended period. Reliable isotopic ages on these dykes are lacking but alkaline plutonic activity is known to have persisted to 35 Ma (Rb-Sr age of the Kialineq intrusion, Brown *et al.* 1977). This period of extensional tectonics during early spreading seems not to have been associated with significant uplift, since the early Oligocene clastic sediments of Kap Dalton and Savoia Halvø (Birkenmejer 1972; Soper & Costa 1976) are free from basaltic detritus. Major uplift took place subsequently along coast-parallel faults which are post-magmatic and carry carbonate mineralization.

The present coastline of central E Greenland is thus marked by a major seaward facing step in the upper crust, developed at two distinct periods in the evolution of this Atlantic-type margin. We have shown that the THOL 1 dyke swarm was associated with the first phase of differential vertical displacement and it cannot therefore represent an early position of the spreading axis, as has been suggested. We propose that the step marks the junction between the normal thickness continental crust of Greenland and crust

modified during a pre-rifting phase of thinning and extension. Further S, a strip of thinned continental crust is inferred to exist between the position of anomaly 24 and the continental rise (Featherstone *et al.* 1977). Beneath the Denmark Strait, at unknown distance offshore from the Blosseville Kyst, thinned continental crust with its basalt cover is likely to pass into abnormally thick oceanic-type crust comparable to that forming the Iceland–Faeroe ridge.

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